DAY 05 · THEORY HANDOUT

Linear Regression: Predict a Number

Predict a number from an input and measure the size of its errors.

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A feature and a target

A test score is a number. Study hours are one possible input for predicting it.

Study hours: feature x	Test score: target y
1	42
2	48
3	64
4	70

Example records. Each row contains one input and one known answer.

A **feature** is an input fact. Here it is study hours, written **x**. A **target** is the known answer. Here it is the test score, written **y**.

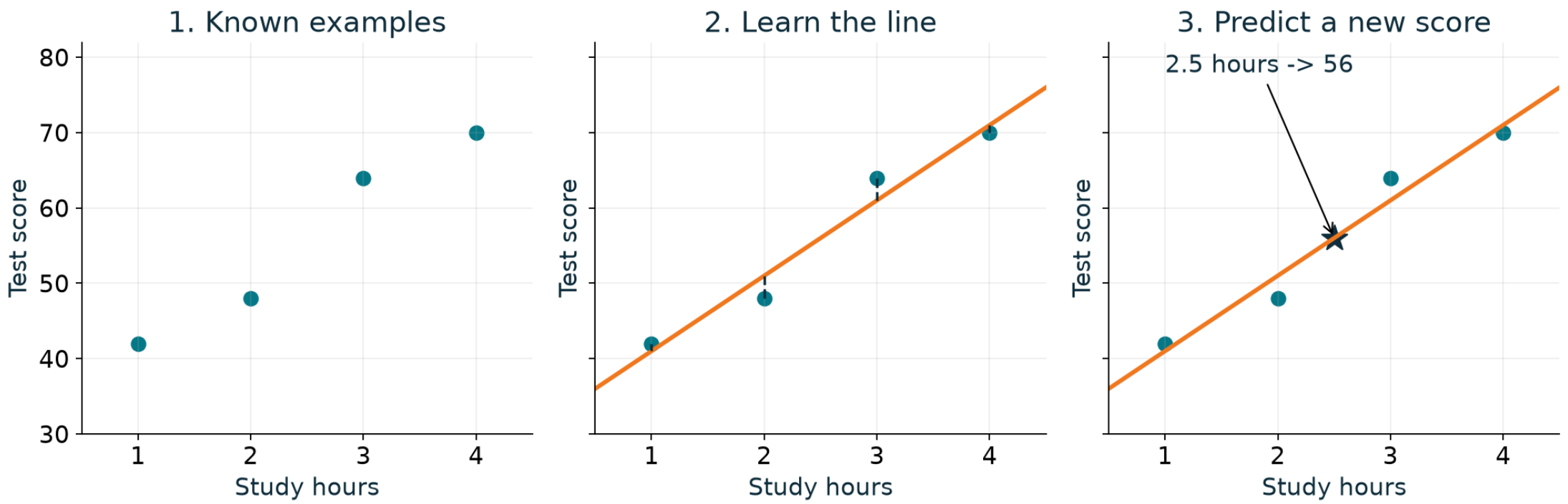
Simple Linear Regression uses one input and predicts a number with a straight line. Multiple Linear Regression uses more than one input.

Each training row has both the input and the real answer. The model learns a relation between them.

> A pattern in data does not prove that study hours alone caused a score.

From known scores to a prediction

A new input uses the same fitted line as the known examples.

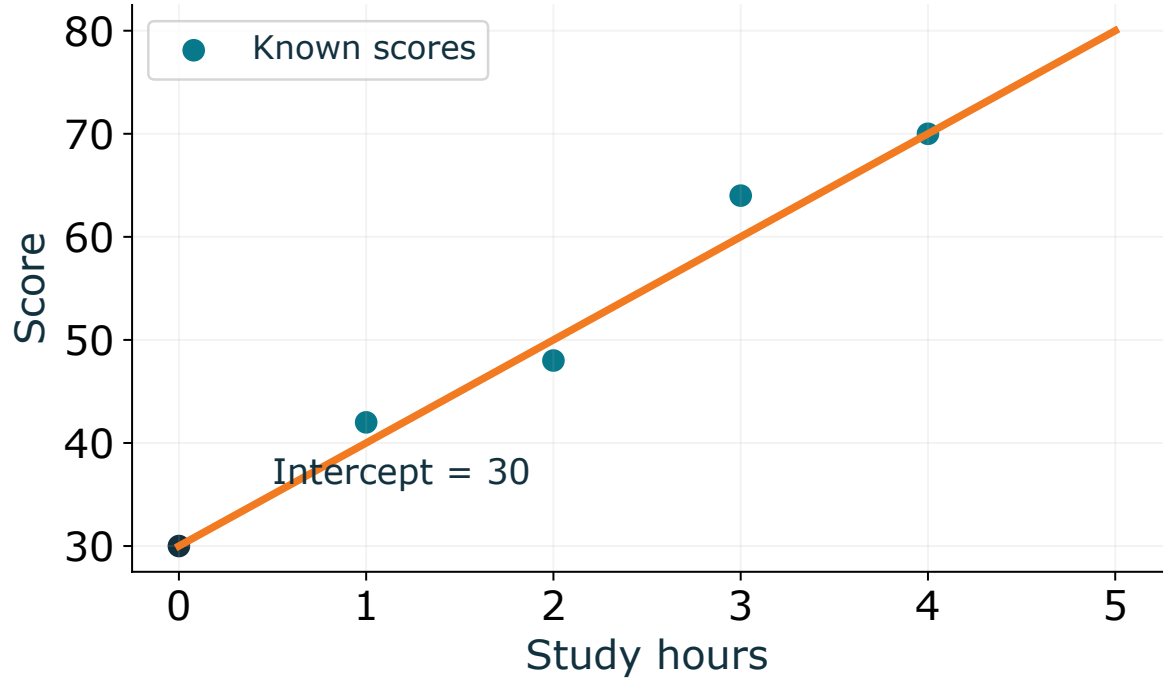


The dashed gaps are residuals. The fitted line predicts **56** at **2.5 study hours**.

> The line is a pattern from these four records. It is not a promise about a new student.

Step 1: start with a candidate line

A candidate model is predicted score = 30 + 10 x study hours.



Intercept = 30: the predicted score when the input is 0. It moves the line up or down.

Slope = 10: one extra study hour adds 10 to the predicted score. It changes how steep the line is.

These are candidate values. Training compares the line predictions with known scores and finds better values.

Slope and intercept are model parameters: values learned from training data.

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Step 2: make predictions

Put each study-hour value into the same candidate line.

Input x	Line calculation	Predicted score
1	30 + 10 x 1	40
2	30 + 10 x 2	50
3	30 + 10 x 3	60
4	30 + 10 x 4	70

For 1 hour, the prediction is 30 + 10 x 1 = **40**. For 3 hours, it is 30 + 10 x 3 = **60**.

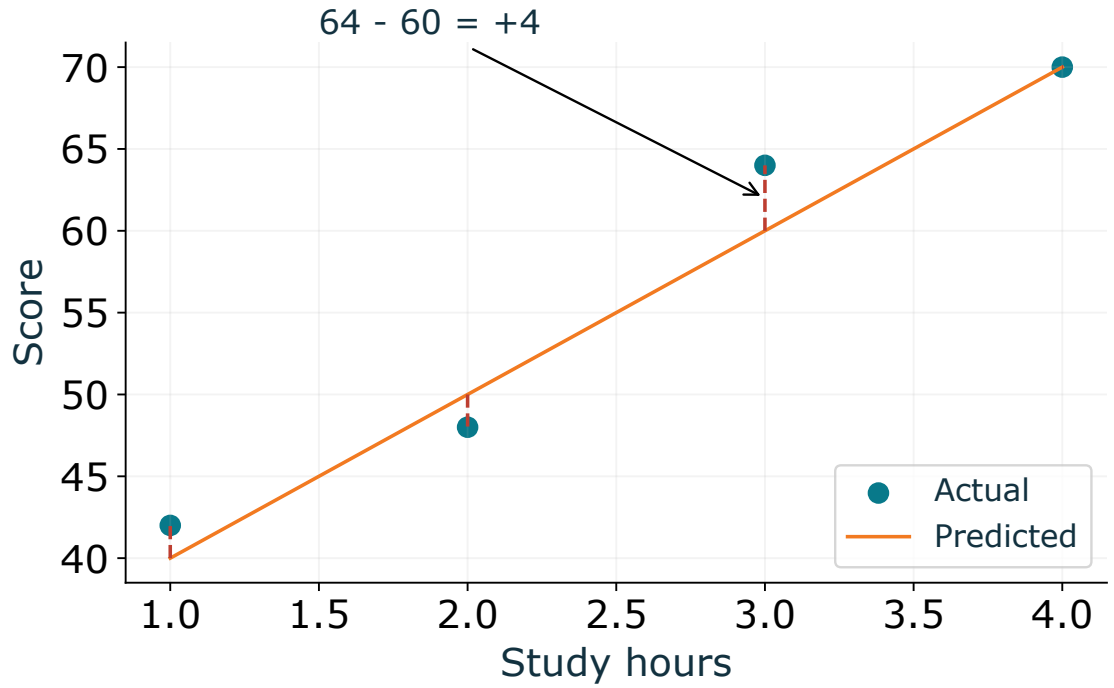
A **prediction** is the model answer. The **actual value** is the answer recorded in the data.

The same line is used for every row. Predictions and actual values must refer to the same row before comparing them.

> The multiplication sign in 30 + 10 x 3 means 10 multiplied by 3.

Step 3: find the error

Compare each prediction with the actual score. The vertical gap is the residual.



Residual = actual minus predicted. At 3 hours, the actual score is 64 and the prediction is 60. The residual is +4.

The four residuals are **+2, -2, +4, 0**. Positive means the model predicted too low. Negative means it predicted too high.

Adding signed errors can hide mistakes: +2 and -2 cancel. Error size or squared error avoids that problem.

A zero residual means that one prediction exactly matches its actual value.

Step 4: compare the lines

Square each residual, then add the squared values for each candidate line.

Candidate	Predictions	Squared errors	Total
A: 35 + 5x	40, 45, 50, 55	4 + 9 + 196 + 225	434
B: 30 + 10x	40, 50, 60, 70	4 + 4 + 16 + 0	24
C: 31 + 10x	41, 51, 61, 71	1 + 9 + 9 + 1	20

All three candidates are checked against the same four known scores.

Squared error means error multiplied by itself. A residual of -3 gives (-3) x (-3) = 9.

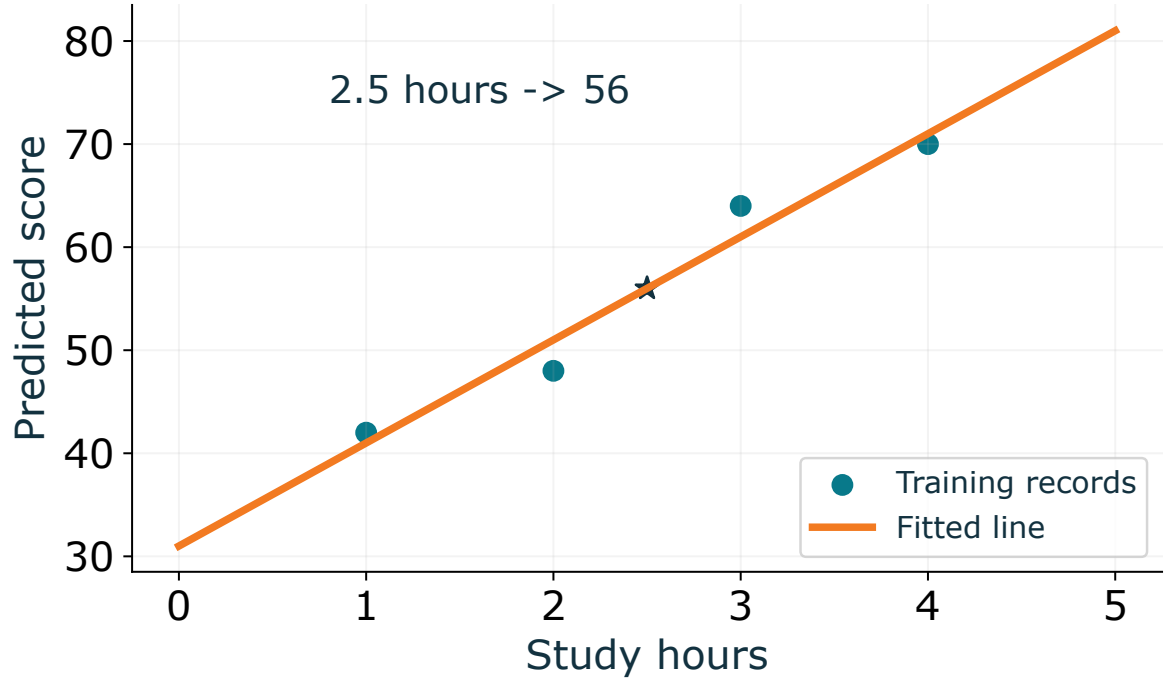
Candidate C has less total squared error than A and B. It moves the intercept from 30 to 31 while keeping slope 10.

Ordinary Least Squares finds the slope and intercept with the smallest total squared error. The computer solves this fit.

Candidate C is the fitted line for these four records. Larger errors count more when squared.

Step 5: predict a new value

The fitted line is predicted score = 31 + 10 x study hours.



For a new input of **2.5 hours**, the model gives 31 + 10 x 2.5 = **56**.

The fitted slope and intercept stay the same during prediction. A new input is not a new training step.

A prediction is an estimate. The real score can be different because other facts also affect results.

A good fit should also work on new records, rather than only the training records.

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Training data and test data

Keep some records separate so the final check uses answers the model did not learn from.



Training data is used to learn the model. **Test data** is held back to check predictions after training.

Do not use test answers to fit the line or choose its settings. That would leak information into the model.

Generalization means working well on new records. **Overfitting** means following training details that do not carry over.

A training error can be small even when the error on new data is large.

MAE: the usual error size

A delivery-time model makes three predictions. How many minutes off is it, on average?

Actual delivery time	Predicted time	Error size
20 minutes	18 minutes	2 minutes
30 minutes	33 minutes	3 minutes
40 minutes	45 minutes	5 minutes

MAE = (2 + 3 + 5) / 3 = 3.33 minutes

Mean Absolute Error (MAE) is the average of error sizes. Ignore the signs, add the sizes, then divide by the number of predictions.

Use MAE when a clear average error in the original unit is useful, such as minutes, money, or marks.

Here MAE is 3.33 minutes. It describes average error size, not a guarantee that every prediction is within 3.33 minutes.

> Lower MAE is better when comparing models on the same data and target unit.

MSE and RMSE: large mistakes count more

Two models can have the same MAE but very different large errors.

Model	Four error sizes	MAE	MSE	RMSE
A	3, 3, 3, 3	3	9	3
B	0, 0, 0, 12	3	36	6

Errors are in minutes. Both models have the same MAE. Model B has one much larger mistake.

Mean Squared Error (MSE) averages squared errors. Squaring gives a large mistake much more weight. MSE is used in fitting and comparisons.

Root Mean Squared Error (RMSE) is the square root of MSE. It returns the result to the original unit.

Use RMSE when a few very large errors deserve extra attention, such as very wrong delivery-time estimates.

Model B has a 12-minute miss. RMSE highlights it even though both models have MAE = 3 minutes.

R-squared: beat the mean guess

A mean guess predicts 30 minutes for every delivery. Compare the model with that simple answer.

Actual minutes	Mean guess	Model prediction
20	30	22
30	30	28
40	30	38

R2 = 1 - 12 / 200 = 0.94

Mean guess squared errors: 100 + 0 + 100 = 200
Model squared errors: 4 + 4 + 4 = 12

R-squared (R2) compares model squared errors with errors from the mean of the test targets.

Use R2 to see whether the model does better than that mean guess on the same test data. Read it beside MAE or RMSE.

1 means perfect predictions. **0** means equal squared error to the mean guess. A negative value means worse.

R2 = 0.94 does not mean 94% of predictions are correct. Regression does not use class accuracy.

Which regression metric should be used?

Choose a metric by asking what kind of error matters for the task.

Metric	Use it when	Example
MAE	Usual error size matters	How far is a price estimate off?
RMSE	Large errors need more weight	Avoid a few very wrong delivery times
MSE	Comparing squared error during fitting	Choose a line with smaller squared gaps
R2	Comparing with a mean guess	Does the model beat one fixed guess?

For a price estimate, MAE gives an easy answer in money: how far off is a prediction on average?

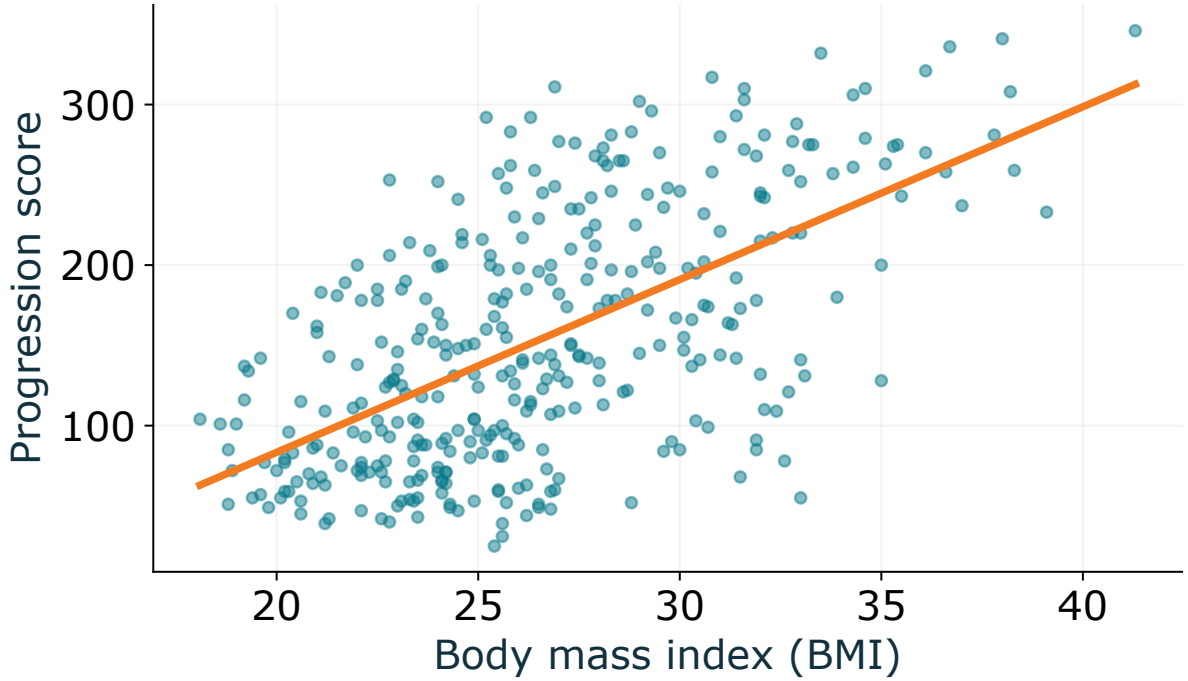
For delivery times, RMSE gives extra weight to a few large mistakes. Check the individual errors too.

R2 gives a comparison with the mean guess. It does not show the error in money or minutes. Different datasets can have very different R2 values.

> Compare models on the same held-out records. One metric alone does not explain every mistake.

An example with real records

The scikit-learn diabetes dataset has 442 records. BMI is one feature; progression score is the target.



Body mass index (BMI) is a number made from height and weight. The target records disease progression one year later.

The example uses 353 training records and 89 test records. The orange line is fitted using the training records only.

Scores vary widely at the same BMI. One feature cannot explain every outcome. Other inputs may carry useful information.

Each dot is a known record. The line shows the model prediction, not a guaranteed result.

Read the test results

Evaluate the fitted model on the 89 held-out records.

Metric	Test result	Meaning
MAE	52.26	score points
MSE	4061.83	squared score points
RMSE	63.73	score points
R2	0.233	compared with the test mean

MAE = 52.26 score points. This is the average absolute error on the test records.

RMSE = 63.73 score points. Its larger value shows that large errors receive more weight.

R2 = 0.233. The model beats the test mean guess, but scores still differ in ways the line does not capture.

> The unit, dataset, and held-out records matter when deciding whether an error is acceptable.

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